

– SQL TEST AUG 27
– ONLINE BATCH 15
– NAME : ARPIT MANOJ BANGRE

-- 1.Find employees whose latest salary increased by more than 30% over their previous salary.

```
with salary_cte as
(
select emp_id,salary,salary_date,
lag(salary) over(partition by emp_id order by salary_date) as previous_salary,
row_number() over(partition by emp_id order by salary_date desc) as rn
from employee_salary
)
select emp_id
from salary_cte
where rn=1 and salary > previous_salary*1.30;
```

-- 5.Business wants top 5 customers based on total lifetime spend.

-- Tables:

-- customers(customer_id, customer_name, signup_date)

-- orders(order_id, customer_id, order_date, amount)

-- Return columns

-- 1.customer_name

-- 2.total_spent

-- 3.rank

```
with spend_cte as
(
select c.customer_name,
sum(o.amount) as total_spent
from customers c join orders o
on c.customer_id=o.customer_id
group by c.customer_name
),
rank_cte as
(
select customer_name,total_spent,
rank() over(order by total_spent desc) as [rank]
from spend_cte
)
select top 5 customer_name,total_spent,[rank]
from rank_cte
order by [rank];
```

-- 6.Identify Peak Sales Hour

-- Scenario:
-- Find hour of day with highest total revenue.
-- Table:
-- sales(order_id, order_time, amount)

```
with hour_cte as
(
select datepart(hour,order_time) as sales_hour,
sum(amount) as total_revenue
from sales
group by datepart(hour,order_time)
)
select top 1 sales_hour,total_revenue
from hour_cte
order by total_revenue desc;
```

-- 9.Find Duplicate Email IDs

```
with email_cte as
(
select email,count(*) as total
from customers
group by email
)
select email,total
from email_cte
where total>1;
```

-- 10. To detect fraud in bank, find Customers doing transactions from different cities at same time

```
with fraud_cte as
(
select t1.customer_id,t1.transaction_time,t1.city as city1,t2.city as city2
from transactions t1 join transactions t2
on t1.customer_id=t2.customer_id
and t1.transaction_time=t2.transaction_time
and t1.city<>t2.city
)
select customer_id,transaction_time,city1,city2
from fraud_cte;
```